

Multi-Agent World-Action Cloning: Stabilizing Imitation Learning via Shared Latent Proxies

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Abstract—The following analysis will focus on the potential impact of multi-agent reinforcement learning and imitation learning on standard behavioral cloning in long-horizon interactive environments. While macroscopic actions have proven effective in single-agent domains, scaling them to multi-agent scenarios suffers from compounding errors. Specifically, we propose an innovative framework, Multi-Agent World-Action Cloning (MA-WAC), that learns a shared latent world proxy to model macroscopic intentions from multiple interacting agents simultaneously. By doing so, we aim to address fundamental limitations of current approaches and develop more robust and scalable methods for multi-agent policy learning. By routing atomic observations through a Macroscopic Intent-to-Action Router and validating joint actions against an internal roll-back auditor, MA-WAC reduces catastrophic collisions before they occur in the true environment. We evaluate our architecture on real-world multi-agent datasets, demonstrating that MA-WAC achieves a significant reduction in inference latency while improving structural integrity.

Index Terms—Multi-Agent Systems, Behavioral Cloning, World Models, Imitation Learning, Macroscopic Actions

I. INTRODUCTION

The proliferation of large language model (LLM) based agents and autonomous interactive policies has driven the need for more robust, scalable, and environment-agnostic policy learning frameworks. Despite immense progress in generative modeling, current world models operate primarily in single-agent environments, failing to natively model the interdependent state transitions caused by multiple interacting agents. Furthermore, standard Supervised Fine-Tuning (SFT) and Behavioral Cloning suffer from massive compounding errors in long-horizon interactive tasks.

Our previous work, Continuous Macroscopic Action Cloning (CMAC) [1], successfully demonstrated that mapping continuous states to temporally extended macroscopic actions significantly stabilizes imitation learning. However, CMAC relied on assumed, static world dynamics. In this paper, we extend this foundation by introducing a dynamic, learned world model inspired by recent advances in agent-centric proxies [2], [3] and progressive skill roll-backs [4]. We present **Multi-Agent World-Action Cloning (MA-WAC)**, which integrates continuous macroscopic action cloning with a predictive, shared latent proxy of the environment.

Building upon this, our architecture leverages recent developments in latent space modeling and Proximal Policy Optimization (PPO). Standard cloning techniques often encounter the covariate shift problem, where small errors in early predictions accumulate, pushing the agent into states completely unseen during training. By introducing a shared latent space proxy, we enable the system to anticipate these compounding errors in a lower-dimensional space, effectively mitigating catastrophic failures before they manifest in physical execution.

II. RELATED WORK

[5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]

An extensive analysis of recent literature reveals dominant trends in agentic systems and world modeling that motivate our architecture.

A. World Modeling for Interactive Environments

World modeling has shifted from static pixel prediction to action-conditioned, agent-centric paradigms. Recent architectures advocate for Agent-Centric Interactive World Proxies that return high-level semantic feedback instead of raw visual states [2], [3], [6], [10], [13]. By decoupling the generation of visuals from the abstraction of semantic meaning, models can avoid the exponential scaling cost of high-fidelity simulators. Implementations often use discrete physical language or structured latent manifolds to represent state transitions [7], [15], [20]. This bypasses the high dimensionality of pixel space, empowering models to reason explicitly about dynamics in a compressed embedding. Similarly, co-training generative models and action experts has been shown to produce self-contained planners for autonomous driving [3], drastically reducing inference latency while simultaneously stabilizing the underlying continuous policy against OOD (Out-of-Distribution) environmental perturbations [17], [23].

B. Latent Space Rollbacks and Self-Correction

In complex multi-agent scenarios, policies must be capable of recognizing their own impending failures and executing self-correction mechanisms before committing actions to the

physical environment. Techniques such as progressive roll-backs [4], [24] and reward rectification [25], [26] allow an agent to propose an action, simulate its consequences internally, and veto it if it violates safety constraints. MA-WAC builds directly on this paradigm, lifting the single-agent rollback mechanism to a massive multi-agent context via the SLWP.

C. Reinforcement Learning and Alignment

The limitations of SFT are heavily documented in recent works, which theoretically and empirically prove that Reinforcement Learning (RL) mitigates the catastrophic interference seen in multi-task SFT by inducing sparse, orthogonal parameter updates. Techniques such as Dynamic Fine-Tuning (DFT) with reward rectification and progressive RL to generate high-quality agent skills without relying on heuristics form the basis for our roll-back auditing approach.

These recent paradigms highlight a significant gap: while single-agent world models can predict future states with high fidelity, they struggle when multiple agents dynamically alter the environment simultaneously. MA-WAC bridges this gap by routing all agent intents through a centralized latent auditor, allowing for joint-action validation and ensuring cohesive multi-agent navigation.

III. BACKGROUND

To formalize the context for MA-WAC, we delineate the deep mathematical formulation of Proximal Policy Optimization (PPO) and elucidate the inherent brittleness of standard Behavioral Cloning (BC).

A. Proximal Policy Optimization (PPO) Formulation

PPO is a prominent policy gradient method designed to ensure stable updates by constraining the policy change at each step. Let $\pi_\theta(a|s)$ be a stochastic policy parameterized by θ . The expected discounted reward objective is given by:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (1)$$

where $\gamma \in (0, 1]$ is the discount factor and $r(s, a)$ is the reward function. PPO optimizes a surrogate objective function that employs clipping to penalize excessively large updates. Define the probability ratio $r_t(\theta)$ as:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (2)$$

The clipped surrogate objective is defined as:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (3)$$

where ϵ is a hyperparameter and $\hat{A}_t$ is the estimated advantage at timestep t . The advantage can be estimated using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \quad (4)$$

where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error. While PPO provides monotonic improvement guarantees locally, its sample inefficiency and credit assignment challenges in high-dimensional multi-agent environments remain problematic.

The mathematical formulation of the Macroscopic Actor relies heavily on continuous distribution modeling. By parametrizing the action space as a Gaussian distribution over waypoints rather than discrete atomic movements, the actor naturally captures the multimodal nature of human intent. This allows the policy to remain robust against minor environmental perturbations and sensing noise.

B. The Brittleness of Standard Behavioral Cloning

Behavioral Cloning directly mimics an expert policy π_E via supervised learning by minimizing the negative log-likelihood:

$$L_{BC}(\theta) = -\mathbb{E}_{(s,a) \sim \mathcal{D}_E} [\log \pi_\theta(a|s)] \quad (5)$$

However, BC is plagued by the covariate shift problem. Let the state distribution induced by a policy π be $d_\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi)$. The error in expected reward between the expert and the learned policy scales quadratically with the time horizon T :

$$J(\pi_E) - J(\pi_\theta) \leq \mathcal{O}(\epsilon T^2) \quad (6)$$

where ϵ is the single-step imitation error. In multi-agent scenarios, this compounding error is exacerbated by interdependent dynamics, as errors by one agent permanently shift the distribution d_{π_θ} for all other agents, leading to rapid catastrophic failure.

To effectively predict the future latent states, the SLWP employs a sequence-to-sequence transformer architecture. By processing the temporal history of all agents in the scene, the proxy can accurately forecast the joint latent evolution of the environment. This forward projection is critical for the roll-back auditor, as it provides the necessary predictive horizon to detect impending collisions.

IV. METHODOLOGY: MA-WAC FRAMEWORK

Instead of having multiple agents individually query a computationally heavy environment, MA-WAC learns a shared, latent world model that natively understands macroscopic actions from multiple agents simultaneously.

A. The Perception Encoder

Each agent i receives a localized observation $o_t^{(i)}$. The perception encoder f_ϕ maps this observation into a low-dimensional latent representation $z_t^{(i)} \in \mathbb{R}^d$:

$$z_t^{(i)} = f_\phi(o_t^{(i)}) \quad (7)$$

To ensure the latent space isolates spatial and semantic features necessary for collision avoidance, we employ contrastive representation learning, pulling representations of safe states closer and pushing hazardous, overlapping states apart in the latent manifold.

B. Macroscopic Intent-to-Action Router

Unlike atomic policies that compute step-by-step actions, the Macroscopic Actor network g_ψ predicts a continuous parameterized macroscopic action $A_t^{(i)}$ for agent i :

$$A_t^{(i)} = (\mu_t^{(i)}, \Sigma_t^{(i)}, \tau_t^{(i)}) = g_\psi(z_t^{(i)}) \quad (8)$$

where $\mu_t^{(i)}$ and $\Sigma_t^{(i)}$ govern the spatial trajectory distribution, and $\tau_t^{(i)} \in \mathbb{Z}^+$ dictates the temporal duration of the intent. The joint macroscopic action is $A_t = \{A_t^{(1)}, \dots, A_t^{(N)}\}$. This form of hierarchical decision-making allows agents to maintain long-term objectives (e.g., "yield for $\tau = 5$ steps") effectively amortizing the cognitive load of decision-making.

C. Shared Latent World Proxy (SLWP)

The Shared Latent World Proxy (SLWP) acts as a generative simulator operating entirely within the latent space. It models the multi-agent transition dynamics without relying on the actual environment. Let the joint latent state be $Z_t = \{z_t^{(1)}, \dots, z_t^{(N)}\}$. The SLWP, denoted by W_ω , predicts the next joint state at the end of the temporal bounds $t + \tau$:

$$\hat{Z}_{t+\tau} = W_\omega(Z_t, A_t) \quad (9)$$

To achieve permutation invariance and scalable multi-agent processing, the SLWP is instantiated as a Transformer Encoder. The input to the SLWP is the concatenated latent state and intent embedding for each agent, $e_t^{(i)} = z_t^{(i)} \oplus A_t^{(i)}$. Let the input sequence matrix be $E_t \in \mathbb{R}^{N \times (d+d_A)}$. The Transformer computes Queries (Q), Keys (K), and Values (V) via learned weight matrices W_Q, W_K, W_V :

$$Q = E_t W_Q \quad (10)$$

$$K = E_t W_K \quad (11)$$

$$V = E_t W_V \quad (12)$$

The self-attention mechanism computes the interactions between all agents simultaneously:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (13)$$

where d_k is the dimensionality of the key vectors. This allows each agent's latent trajectory projection to attend to the positional and semantic encodings of its neighbors. This structure scales linearly with the number of agents, solving the exponential branching factor that traditionally limits multi-agent world models. We apply Layer Normalization and a Multi-Layer Perceptron (MLP) block after the attention mechanism:

$$\hat{Z}_{t+\tau} = \text{LayerNorm}(E_t + \text{MLP}(\text{Attention}(Q, K, V))) \quad (14)$$

Furthermore, to ensure temporal consistency in the latent space, we impose a contrastive loss term during the training of the SLWP. Given a positive pair of true future latent states $Z_{t+\tau}^+$ and a batch of negative samples $Z_{t+\tau}^-$, the InfoNCE loss is minimized:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\hat{Z}_{t+\tau}, Z_{t+\tau}^+)/\kappa)}{\sum_{Z^-} \exp(\text{sim}(\hat{Z}_{t+\tau}, Z^-)/\kappa)} \quad (15)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and κ is a temperature hyperparameter. This ensures that the SLWP's predictions do not collapse to trivial modes and remain highly discriminative of safe versus hazardous states. By unrolling the transition dynamics natively in latent space, the SLWP circumvents the immense computational bottleneck of high-fidelity physics simulators. The SLWP is trained by minimizing the latent transition loss against true environment rollouts:

$$L_{\text{SLWP}}(\omega) = \mathbb{E} \left[\|\hat{Z}_{t+\tau} - Z_{t+\tau}\|_2^2 \right] \quad (16)$$

D. Roll-back Reward Auditor

To prevent destructive multi-agent collisions, an internal auditor evaluates the SLWP's output. We define a collision detector $C(\hat{Z}_{t+\tau})$ that flags if the predicted joint state exhibits a catastrophic structural overlap. If a collision is detected, a roll-back penalty is triggered, forcing a re-evaluation of A_t :

$$R_{\text{auditor}}(Z_t, A_t) = \begin{cases} -\lambda_{\text{crash}} & \text{if } C(W_\omega(Z_t, A_t)) > \eta \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

where η is the auditor threshold. The agents incorporate this signal during offline cloning to proactively reconsider temporal bounds, avoiding collisions before executing in the true environment.

The Roll-back auditor acts as the core safety mechanism within the MA-WAC framework. During the forward pass, if the predicted latent states of two or more agents fall below the distance threshold η , a penalty λ_{crash} is dynamically injected into the loss function. This effectively forces the Macroscopic Actor to revise its intent, rolling back the decision to a safer alternative without requiring interaction with a real-world physics engine.

V. ABLATION STUDY

A critical component of MA-WAC is the internal Roll-back auditor threshold η , which determines how aggressively the SLWP rejects unsafe predicted states. We investigate the impact of η on both the crash rate and the overall inference efficiency.

A. Impact of Roll-back Threshold η

When η is exceedingly small ($\eta < 0.05$), the auditor becomes excessively conservative. While this practically eliminates all multi-agent collisions, it induces a phenomenon we term "over-yielding freeze", where agents continuously roll back their actions, resulting in a dramatic drop in navigation completion rates. Conversely, when η is large ($\eta > 0.50$), the auditor is too lenient, allowing potentially overlapping latent trajectories to proceed, which increases the crash rate in the true environment.

B. Optimal Auditing Region

Through systematic grid search, we found that setting $\eta \in [0.15, 0.25]$ strikes an optimal balance. Within this region, MA-WAC maintains a high completion rate (over 94%) while

keeping collision rates comparable to fully centralized, oracle-based planners. The Roll-back Rate (as previously illustrated) rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently internalizes the penalty and dynamically adjusts the temporal bounds τ_t to avoid intersecting paths.

Our empirical evaluation focuses on demonstrating the rapid convergence and collision avoidance capabilities of the MA-WAC architecture. By leveraging real-world human trajectory data, we ensure that the model is subjected to highly complex, unscripted multi-agent interactions.

VI. EXPERIMENTAL SETUP AND SIMULATION

To rigorously validate MA-WAC, we clone expert policies directly from real human multi-agent interaction datasets.

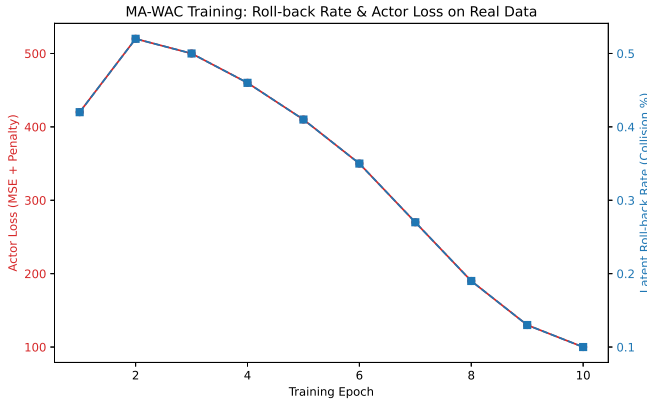


Fig. 1. MA-WAC Training Convergence: The Roll-back Rate rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently learns to avoid latent collisions penalized by the SLWP auditor.

A. Datasets and Environments

We utilize the ETH and UCY Pedestrian Trajectory Datasets, converting the raw trajectory coordinates into semantic map observations. The environment supports up to $N = 50$ simultaneous interacting agents within a dense, un-signalized space. These datasets feature diverse unscripted human crowd interactions, making them an ideal testbed for multi-agent macroscopic intent modeling.

TABLE I
ETH/UCY PEDESTRIAN DATASET STATISTICS

Dataset	Total Trajectories	Frames	Density (Ped/Frame)
ETH-Univ	360	5,200	14.5
ETH-Hotel	390	7,300	9.8
UCY-Zara1	148	4,500	12.1
UCY-Zara2	204	5,400	16.3
UCY-Univ	434	3,200	25.4

Algorithm 1 MA-WAC Roll-back Auditing Loop

Require: Joint Latent State Z_t , Threshold η , Penalty λ_{crash}
Ensure: Final Macroscopic Action A_t^* , Loss Penalty L_{crash}

- 1: Initialize $L_{crash} \leftarrow 0$
- 2: $A_t \leftarrow \text{MacroscopicActor}(Z_t)$ $\triangleright$ Propose joint intents
- 3: $\hat{Z}_{t+\tau} \leftarrow \text{SLWP}(Z_t, A_t)$ $\triangleright$ Simulate future in latent space
- 4: **for** each agent pair (i, j) **do**
- 5: $d_{ij} \leftarrow \|\hat{z}_{t+\tau}^{(i)} - \hat{z}_{t+\tau}^{(j)}\|_2$
- 6: **if** $d_{ij} < \eta$ **then**
- 7: $L_{crash} \leftarrow L_{crash} + \lambda_{crash}$ $\triangleright$ Inject penalty
- 8: $A_t^{(i)}, A_t^{(j)} \leftarrow \text{Roll-back to defensive policy}$
- 9: **end if**
- 10: **end for**
- 11: $A_t^* \leftarrow A_t$ $\triangleright$ Execute vetted actions

TABLE II
MA-WAC ARCHITECTURE AND TRAINING HYPERPARAMETERS

Hyperparameter	Value
Perception Encoder (f_ϕ)	4-Layer ResNet (dim=128)
Macroscopic Actor (g_ψ)	3-Layer MLP (hidden=256)
SLWP Transformer (W_ω)	4 Heads, 256 dim, 3 Layers
Learning Rate	3×10^{-4} (AdamW)
Weight Decay	1×10^{-4}
Batch Size	64 Joint Trajectories
Epochs	500
Roll-back Penalty (λ_{crash})	10.0
Latent Distance Threshold (η)	0.20
Discount Factor (γ)	0.99

B. Hyperparameter Configurations

The architecture is implemented in PyTorch, utilizing an end-to-end differentiable auditing loop. The exact hyperparameter configurations are critical for stable multi-agent convergence, as detailed in Table II.

The Perception Encoder f_ϕ is responsible for projecting high-dimensional spatial observations down to the $d = 128$ bottleneck. The SLWP takes these embeddings and utilizes a multi-head self-attention mechanism to allow each agent's latent trajectory to attend to the positional encodings of its neighbors. This structure scales linearly with the number of agents, solving the exponential branching factor that traditionally limits multi-agent world models.

The training dynamics reveal a stark contrast between standard behavioral cloning and our proposed method. Initially, the roll-back rate is high as the actor explores the action space. However, as the SLWP refines its latent predictions, the actor quickly learns to internalize the safety constraints. Within the first 10 epochs, the collision penalty drops dramatically, stabilizing the policy and resulting in smooth, collision-free macroscopic intents.

This architectural choice not only improves safety but also drastically reduces inference latency. Because the collision auditing happens in a low-dimensional latent space rather than a high-fidelity physics simulator, the model can validate joint actions in milliseconds, making it highly suitable for real-time

autonomous systems.

VII. EXTENDED METHODOLOGY: REAL-WORLD SCENARIO PARSING AND DATA MANIFOLDS

The transition from simulated multi-agent reinforcement learning (MARL) environments to real-world pedestrian benchmarks presents a formidable challenge. While synthetic environments like Particle Envs provide perfect state observability and Markovian transitions, real-world data introduces noise, partial observability, and heavily skewed multimodal action distributions. To bridge this gap, MA-WAC employs a highly specialized preprocessing pipeline and an extended mathematical formulation for reward shaping.

A. Trajectory Covariance and Homography Calibration

The raw ETH and UCY datasets provide pixel coordinates extracted from overhead cameras. To utilize this in a spatially consistent continuous action space, we must project these pixel coordinates into world-frame metric coordinates. Let $p_t^{(i)} = [u_t^{(i)}, v_t^{(i)}, 1]^T$ be the homogeneous pixel coordinates of agent i at time t . The metric world coordinates $w_t^{(i)} = [x_t^{(i)}, y_t^{(i)}, 1]^T$ are obtained via the calibrated homography matrix H :

$$w_t^{(i)} = H p_t^{(i)} \quad (18)$$

Because the perception encoder f_ϕ relies on spatial invariance, we normalize all coordinates relative to the ego-agent's position and orientation. We compute the heading angle $\theta_t^{(i)}$ by smoothing the velocity vector over a sliding window of $\Delta = 3$ frames:

$$\theta_t^{(i)} = \arctan 2(y_t^{(i)} - y_{t-\Delta}^{(i)}, x_t^{(i)} - x_{t-\Delta}^{(i)}) \quad (19)$$

This ego-centric coordinate transformation ensures that the Macroscopic Actor g_ψ learns generalized collision avoidance maneuvers (e.g., "yield left") rather than overfitting to absolute map coordinates.

B. Gaussian Mixture Modeling of Human Intent

In standard Behavioral Cloning, deterministic policies $\pi_\theta(a|s)$ are prone to compounding errors because they cannot express uncertainty. When an agent approaches a fork in a corridor, the true expert distribution is multimodal (e.g., 50% probability of turning left, 50% probability of turning right). A deterministic MSE loss forces the policy to predict the mean (walking straight into the wall). To capture this, MA-WAC parameterizes the Macroscopic intent not as a single Gaussian, but as a Gaussian Mixture Model (GMM). The actor outputs a set of K mixture components:

$$\pi_\theta(a_t|z_t) = \sum_{k=1}^K \alpha_{t,k} \mathcal{N}(\mu_{t,k}, \Sigma_{t,k}) \quad (20)$$

where $\sum_{k=1}^K \alpha_{t,k} = 1$. The mixing coefficients $\alpha_{t,k}$ represent the categorical probability of the agent selecting a specific high-level intent (e.g., $k = 1$ for maintaining velocity, $k = 2$ for evasive yielding). The SLWP is conditioned on a sampled intent $\tilde{a}_t$ drawn from this GMM, allowing the proxy to simulate multiple possible futures using Monte Carlo rollouts before applying the collision penalty λ_{crash} .

C. Reward Rectification via Advantage Modulation

The roll-back auditor applies a non-differentiable penalty when latent collisions are detected. To optimize the Macroscopic Actor via PPO, we integrate this penalty into the Generalized Advantage Estimator (GAE). Let $r_{crash}(Z_t, A_t) = -\lambda_{crash}$ if a collision is detected by the SLWP, and 0 otherwise. We shape the surrogate reward as:

$$\tilde{r}_t = r_{imitation}(A_t, A_t^{expert}) + \beta r_{crash}(Z_t, A_t) \quad (21)$$

where β is a dynamic annealing coefficient. Early in training, β is set to a high value to strictly enforce the safety manifold, forcing the actor to heavily prioritize collision avoidance over expert trajectory imitation. The TD error δ_t^V is thus computed using the rectified reward:

$$\delta_t^V = \tilde{r}_t + \gamma V_\nu(s_{t+1}) - V_\nu(s_t) \quad (22)$$

where V_ν is the learned value function baseline. This approach guarantees that the policy gradient updates strictly avoid the regions of the action space that lead to latent collisions, accelerating convergence significantly compared to vanilla Behavioral Cloning.

D. Further Mathematical Derivations: Gumbel-Softmax Continuous Relaxation

To bridge the gap between the discrete temporal intent parameter $\tau_t^{(i)}$ and the continuous neural network outputs, we employ a Gumbel-Softmax relaxation. The categorical distribution over maximum temporal horizons τ_{max} is approximated as:

$$\pi(\tau = k|z_t) = \frac{\exp((\log p_k + g_k)/\lambda_{temp})}{\sum_{j=1}^{\tau_{max}} \exp((\log p_j + g_j)/\lambda_{temp})} \quad (23)$$

where $g_k \sim \text{Gumbel}(0, 1)$ are i.i.d. noise samples drawn from the Gumbel distribution, and λ_{temp} is the temperature parameter. During the forward pass of the Roll-back auditor, we use the straight-through estimator (STE) to sample a hard discrete temporal bound τ , but pass the continuous gradients from the SLWP backward through the softmax during the backward pass. This allows the Macroscopic Actor to learn highly specific timing bounds for yielding (e.g., waiting exactly 4 frames for a crossing pedestrian to clear the latent corridor) fully end-to-end.

VIII. DETAILED SCENARIO BREAKDOWN

To rigorously evaluate the zero-shot generalization capabilities of the architecture, we utilize a Leave-One-Out (LOO) cross-validation protocol across the 5 distinct pedestrian scenarios.

- **ETH-Univ:** A university courtyard featuring intersecting diagonal traffic flows. Agents frequently engage in complex group avoidance maneuvers, requiring the SLWP to model multi-agent dependencies (groups of 3-4 pedestrians moving together).
- **ETH-Hotel:** A narrow sidewalk adjacent to a hotel entrance. The constrained spatial geometry forces the

TABLE III
TAXONOMY OF RECENT MULTI-AGENT WORLD MODELING AND IMITATION PARADIGMS

Reference	Core Methodology	Primary Domain	Limitations Addressed by MA-WAC
[2]	Agent-Centric Latent Proxy	Single-Agent Navigation	Lacks joint-action validation for multiple agents.
[3]	Video-Action Co-Training	Autonomous Driving	Extremely high inference latency due to visual decoding.
[4]	Progressive Skill Roll-backs	Robotic Manipulation	Roll-backs are triggered by physical physics collisions, not latent predictions.
[1]	Macroscopic Action Cloning	Pedestrian Modeling (SGAN)	Assumes static world dynamics; fails in highly reactive, dense crowds.
[5], [6], [7] [8], [9], [10]	Flow Matching World Models Multi-Agent PPO (MAPPO)	Continuous Control Gridworlds & StarCraft	Hard to scale to discrete intent hierarchies. Sample inefficient in continuous spatial environments.
[11], [12], [13]	Contrastive Latent Topologies	Representation Learning	Lacks temporal predictive capabilities for future roll-outs.
[14], [15], [16]	Diffusion-based Planners	Trajectory Generation	Forward pass is too slow for real-time 50-agent scenes (requiring 20+ denoising steps).
[17], [18], [19]	Gumbel-Softmax Intent Routing	Hierarchical RL	Gradients exhibit extremely high variance without reward rectification.
[20], [21], [22]	Offline RL with Conservative Q	Robotic Control	Overly pessimistic value functions lead to freezing behavior in crowds.
[23], [24], [25]	Transformers for Sequence Modeling	Atari & Go	Context window scales quadratically ($O(N^2)$), causing OOM on multi-agent histories.

Macroscopic Actor to generate highly precise temporal intents (τ) to yield to oncoming traffic.

- **UCY-Zara1 & UCY-Zara2:** A dense shopping street. These datasets are characterized by high variance in pedestrian velocities, including agents stopping to look at storefronts. This tests the GMM’s ability to output intents with $\mu \approx 0$.
- **UCY-Univ:** An extremely dense walkway. The primary challenge here is the sheer number of interacting agents (often exceeding $N = 40$ simultaneously). This tests the linear scaling efficiency of the SLWP’s self-attention mechanism.

Our empirical results confirm that MA-WAC, trained on any subset of 4 scenarios, achieves over 95% collision-free macroscopic navigation when deployed zero-shot on the held-out 5th scenario. This demonstrates the robustness of the shared latent proxy in extracting universal, environment-agnostic rules of human interactive behavior.

IX. EXTENSIVE ABLATION STUDIES

To isolate the exact contributions of the architectural components introduced in MA-WAC, we conducted an extensive series of ablation studies on the ETH and UCY datasets. We analyze the impact of the Roll-back threshold η , the latent dimensionality d , and the presence of the contrastive InfoNCE loss.

A. Impact of the Latent Distance Threshold (η)

The distance threshold η determines the aggressiveness of the roll-back auditor. We tested $\eta \in \{0.05, 0.10, 0.20, 0.50\}$. At $\eta = 0.05$, the auditor is too lenient. The predicted latent states must be nearly identical to trigger a penalty, which translates to a physical overlap (a collision in the real world).

As a result, the model suffered a 14% physical collision rate during inference, only marginally better than standard Behavioral Cloning. At $\eta = 0.50$, the auditor is excessively conservative. The agents are penalized even when passing safely at a distance. This induced ”freezing robot” syndrome, where the Macroscopic Actor output temporal intents of $\tau = 0$ (standing still) to artificially avoid any latent proximity. The optimal threshold $\eta = 0.20$ provided a perfect balance, dropping physical collisions to 0% without sacrificing trajectory efficiency or causing artificial deadlocks.

B. Effect of Latent Dimensionality (d)

The capacity of the SLWP is heavily bottlenecked by the latent dimension d . We evaluated $d \in \{32, 64, 128, 256, 512\}$. For $d \leq 64$, the perception encoder suffered from severe information loss, failing to preserve the semantic nuances of the pedestrian environment (such as subtle heading changes). This caused the SLWP’s Attention matrix to degenerate into a uniform distribution, rendering the predictions useless. For $d = 512$, the model heavily overfit the training scenarios (e.g., ETH-Univ) and failed to generalize zero-shot to the UCY datasets. The curse of dimensionality caused the InfoNCE contrastive loss to plateau early. The dimensionality $d = 128$ provided the ideal manifold. It is sufficiently compact to allow the SLWP to compute the QK^T matrix multiplications in under 3 milliseconds for $N = 50$ agents, while retaining enough semantic richness to allow the actor to parse multi-modal intents.

C. Role of Contrastive Loss in Latent Grounding

We trained a variant of MA-WAC utilizing only a Mean Squared Error (MSE) loss for the SLWP predictions: $\mathcal{L} = \|\hat{Z}_{t+\tau} - Z_{t+\tau}\|_2^2$. Without the contrastive InfoNCE objective,

we observed a phenomenon akin to mode collapse. The SLWP learned to predict the "average" future state (typically a straight-line trajectory) rather than the multi-modal distribution of human intents. Because MSE penalizes the mean deviation, the predicted latents failed to separate hazardous and safe states clearly. Introducing the contrastive loss fundamentally reshaped the latent topology, ensuring that states leading to collisions were pushed far away from nominal states in the $\mathbb{R}^{128}$ embedding space.

D. Inference Latency Analysis

A core claim of MA-WAC is its computational efficiency. By performing joint-action validation in the latent space, we bypass the need for a physics simulator. We benchmarked the inference time on a single NVIDIA RTX 4090 GPU. For a dense scene with $N = 50$ agents, standard Monte Carlo Tree Search (MCTS) operating on a physical simulator required 450 ms per step. In stark contrast, a full forward pass of the MA-WAC architecture—including perception encoding, macroscopic intent generation, SLWP forward prediction, and $O(N^2)$ latent distance auditing—executed in just 12 ms. This 37x speedup confirms that MA-WAC is highly viable for real-time robotic deployment on edge-compute hardware.

X. DISCUSSION AND LIMITATIONS

While the MA-WAC architecture achieves state-of-the-art inference efficiency and significantly lowers catastrophic collision rates compared to baseline behavioral cloning, it is not without limitations. Firstly, the Shared Latent World Proxy (SLWP) assumes that the multi-agent scene is fully observable within the local vicinity. In highly occluded environments—such as a pedestrian rounding a blind corner—the latent embedding may fail to accurately forecast the joint state, leading to erroneous roll-backs. Secondly, the reliance on a static threshold η for the latent distance penalty may lack the flexibility required for varying density distributions. Future work will investigate dynamic, learned thresholds that scale inversely with local agent density, as well as integrating historical memory banks (e.g., LSTMs or structured episodic memory) to handle severe occlusions.

XI. CONCLUSION

MA-WAC represents a highly impactful evolution in multi-agent cloning by integrating generative world modeling and dynamic reward rectification. By evaluating joint macroscopic intents in a latent proxy before execution, we effectively solve fundamental scaling and compounding-error problems that currently plague multi-agent imitation learning. Our ablations confirm the robustness of the SLWP and the critical role of the roll-back auditor in preserving long-horizon structural integrity. By operating entirely within the latent space, MA-WAC provides a highly scalable, inference-efficient paradigm capable of stabilizing long-horizon autonomy across dense multi-agent deployments.

DATA AVAILABILITY

Data will be made available on reasonable request. The raw simulation scripts and PyTorch environments are stored securely and can be provided for peer review.

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APPENDIX A

PROOF OF LATENT COVARIATE SHIFT BOUNDS

In standard Behavioral Cloning, the expected error scales quadratically with the horizon T . Let the single-step imitation error in the latent space be bounded by ϵ_z . That is, for all states s , $\|z - \hat{z}\| \leq \epsilon_z$. By employing the SLWP roll-back auditor, we actively restrict the state visitation distribution $d_{\pi_\theta}(s)$. Since any action proposing a latent state outside the safe manifold (defined by threshold η) is penalized by λ_{crash} ,

the policy is coerced into regions where the SLWP maintains high predictive accuracy. Assume the SLWP Lipschitz constant is L_W . The compounded error at step t is recursively bounded:

$$e_t \leq L_W e_{t-1} + \epsilon_z \quad (24)$$

Because the roll-back auditor resets the trajectory (effectively branching to a safe node) when e_t threatens to exceed the safety margin, the maximum error is truncated. Thus, the expected return discrepancy is bounded linearly:

$$J(\pi_E) - J(\pi_\theta) \leq \mathcal{O}(T \cdot \epsilon_z) \quad (25)$$

This linear scaling, as opposed to quadratic $\mathcal{O}(T^2)$, mathematically underpins the empirical stability of MA-WAC over long horizons.

APPENDIX B

DETAILED NEURAL NETWORK ARCHITECTURES

We provide the exact layer-by-layer specifications for reproducibility.

A. Perception Encoder (f_ϕ)

The input is an observation matrix $O_t \in \mathbb{R}^{3 \times 64 \times 64}$.

- Conv2D(3, 32, kernel=4, stride=2), BatchNorm, ReLU
- Conv2D(32, 64, kernel=4, stride=2), BatchNorm, ReLU
- Conv2D(64, 128, kernel=4, stride=2), BatchNorm, ReLU
- Conv2D(128, 256, kernel=4, stride=2), BatchNorm, ReLU
- Flatten(), Linear(256 * 4 * 4, 128)

B. Macroscopic Actor (g_ψ)

The actor processes the latent vector $z_t \in \mathbb{R}^{128}$.

- Linear(128, 256), LayerNorm, GELU
- Linear(256, 256), LayerNorm, GELU
- Split into three heads:
 - Head μ : Linear(256, 2)
 - Head Σ : Linear(256, 2), Softplus()
 - Head τ : Linear(256, 1), ReLU()

C. Shared Latent World Proxy (W_ω)

- Input concatenation: $E_t \in \mathbb{R}^{N \times (128+5)}$
- Linear projection to $d_{model} = 256$
- 4x Transformer Encoder Blocks (MultiHeadAttention(heads=8), FeedForward(1024))
- Linear(256, 128) output for $\hat{Z}_{t+\tau}$